R Lab 10. Regression Model Selection (continued)

```
# Started in Lab 9
```

4. Sequential F-tests

```
# Forward and backward selection with partial F-tests at each step.
# This tool is in the package SignifReg
# (significance testing in regression model building)

> install.packages("SignifReg")
> library(SignifReg)

> data(longley)
> names(longley)
[1] "GNP.deflator" "GNP"          "Unemployed"    "Armed.Forces" "Population"
"Year"    "Employed"

# Longley's macroeconomic data set is pre-loaded in R. We'll use it to
predict the unemployment rate.

> null = lm( Unemployed ~ 1, data=longley )
> SignifReg( null )

Call: lm(formula = Unemployed ~ Population + GNP + Year, data = longley)

Coefficients:
(Intercept)   Population          GNP         Year
 -1.840e+05    3.924e+01   -6.599e+00    9.326e+01

# Of all available variables, R selected only the Population, Gross National
Product and Year. By default, each F-test is at the α-to-enter level α =
0.05. With a higher α, more variables will be considered significant and
included into the model. With a lower α, it is harder to overcome a
significance criterion, and the model will contain fewer variables.

> SignifReg( null, alpha=0.2 )

Call:  lm(formula = Unemployed ~ Population + GNP + Year + Armed.Forces +
GNP.deflator, data = longley)

Coefficients:
```

```
 (Intercept)    Population           GNP          Year  Armed.Forces  GNP.deflator
  -1.524e+05     3.510e+01    -6.484e+00     7.686e+01    -2.714e-01     9.649e+00
```

```
> SignifReg( null, alpha=0.001 )

Call:  lm(formula = Unemployed ~ 1, data = longley)

Coefficients:
(Intercept)
      319.3
```

Backward elimination is similar. We can set a desired α-to-remove level.

```
> full = lm( Unemployed ~ ., data=longley )
> SignifReg( full, direction="backward" )

Call:
lm(formula = Unemployed ~ GNP + Population + Year, data = longley)

Coefficients:
(Intercept)          GNP   Population         Year
 -1.840e+05   -6.599e+00    3.924e+01    9.326e+01
```

To see all the steps of this variable selection, use option trace = TRUE

```
> SignifReg( full, direction="backward", trace=TRUE )

Call:  lm(formula = Unemployed ~ ., data = longley)

Coefficients:
 (Intercept)  GNP.deflator           GNP  Armed.Forces    Population          Year      Employed
  -1.655e+05     3.815e+00    -3.397e+00    -4.287e-01     1.044e+01     8.582e+01    -3.244e+01


              RSS         AIC       BIC       adj.rsq max_pvalue alpha_cut-off Bonferroni FDR
<none>        1342.97639  132.28691 138.46762 0.98291 0.25847    FALSE         FALSE      FALSE
- GNP.deflator 1560.0956   132.68465 138.09277 0.98213 0.53545    FALSE         FALSE      FALSE
- GNP          3646.0413   146.26697 151.67509 0.95825 0.39287    FALSE         FALSE      FALSE
- Armed.Forces 5915.34581  154.0095  159.41762 0.93226 0.43263    FALSE         FALSE      FALSE
- Population   1573.97601  132.82638 138.2345  0.98198 0.58404    FALSE         FALSE      FALSE
- Year         11193.49526 164.21402 169.62214 0.87181 0.37876    FALSE         FALSE      FALSE
- Employed     3896.1284   147.32843 152.73655 0.95538 0.06106    FALSE         FALSE      FALSE

Call:
lm(formula = Unemployed ~ GNP.deflator + GNP + Armed.Forces +
    Population + Year, data = longley)

Coefficients:
```

```
 (Intercept)  GNP.deflator           GNP  Armed.Forces    Population          Year
  -1.524e+05     9.649e+00    -6.484e+00    -2.714e-01     3.510e+01     7.686e+01
```

	RSS	AIC	BIC	adj.rsq	max_pvalue	alpha_cut-off	Bonferroni	FDR
<none>	3896.1284	147.32843	152.73655	0.95538	0.06106	FALSE	FALSE	FALSE
- GNP.deflator	5630.44594	151.21972	155.85525	0.94138	0.05403	FALSE	FALSE	FALSE
- GNP	36816.1171	181.26367	185.8992	0.61672	0.59076	FALSE	FALSE	FALSE
- Armed.Forces	6310.78085	153.04485	157.68038	0.9343	0.11314	FALSE	FALSE	FALSE
- Population	9166.33965	159.01731	163.65284	0.90457	0.71433	FALSE	FALSE	FALSE
- Year	12142.87902	163.51658	168.15212	0.87358	0.24054	FALSE	FALSE	FALSE

```
Call:
lm(formula = Unemployed ~ GNP + Armed.Forces + Population + Year, data = longley)

Coefficients:
 (Intercept)           GNP  Armed.Forces    Population          Year
  -1.903e+05    -5.799e+00    -2.694e-01     2.625e+01     9.713e+01
```

	RSS	AIC	BIC	adj.rsq	max_pvalue	alpha_cut-off	Bonferroni	FDR
<none>	5630.44594	151.21972	155.85525	0.94138	0.05403	FALSE	FALSE	FALSE
- GNP	39044.27121	180.20384	184.06678	0.62739	0.72154	FALSE	FALSE	FALSE
- Armed.Forces	8010.88252	154.86151	158.72446	0.92355	0.00083	TRUE	TRUE	TRUE
- Population	9283.92097	157.22124	161.08419	0.9114	0.00209	TRUE	TRUE	TRUE
- Year	25320.2248	173.27435	177.1373	0.75836	0.48746	FALSE	FALSE	FALSE

```
Call:
lm(formula = Unemployed ~ GNP + Population + Year, data = longley)

Coefficients:
(Intercept)           GNP    Population          Year
 -1.840e+05    -6.599e+00     3.924e+01     9.326e+01
```

```
Call:
lm(formula = Unemployed ~ GNP + Population + Year, data = longley)

Coefficients:
(Intercept)           GNP    Population          Year
 -1.840e+05    -6.599e+00     3.924e+01     9.326e+01
```

Since any stepwise variable selection algorithm involves multiple tests, we can control the familywise error rate by using the Bonferroni correction

```
> SignifReg( full, direction="backward", correction="Bonferroni" )
```

Compare results of stepwise variable selection without the Bonferroni correction and with it.

Bonferroni reduces the alpha levels and makes it harder for variables to enter the model.

```
> SignifReg( null, alpha=0.2 )

Call:
lm(formula = Unemployed ~ Population + GNP + Year + Armed.Forces +
    GNP.deflator, data = longley)

Coefficients:
 (Intercept)    Population           GNP          Year  Armed.Forces  GNP.deflator
  -1.524e+05     3.510e+01    -6.484e+00     7.686e+01    -2.714e-01     9.649e+00

> SignifReg( null, alpha=0.2, correction="Bonf" )

Call:
lm(formula = Unemployed ~ Population + GNP + Year, data = longley)

Coefficients:
(Intercept)    Population           GNP          Year
 -1.840e+05     3.924e+01    -6.599e+00     9.326e+01
```

5. Visualization: a scatterplot matrix

Scatterplot matrix is a way to visualize relations between the response and predictor variables. It is used to show

1) whether there is a relation between Y and each X_j,
2) whether this relation is linear # or nonlinear,
3) whether there may be strong multicollinearity.

This particular data set is known for its strong multicollinearity. Therefore, careful variable selection is really necessary here.

```
> par(mfrow=c(7,7))
> plot(longley)
```

